

Research article

Modelling cultural ecosystem services of river landscapes in the Iberian Peninsula with deep learning and social media images

Francesc Comalada^{a,b,*}, Vicenç Acuña^a, Xavier Garcia^a^a Catalan Institute for Water Research (ICRA - CERCA), Carrer Emili Grahit 101, Girona, 17003, Spain^b University of Girona, Plaça de Sant Domènec 3, 17004, Girona, Spain

ARTICLE INFO

Keywords:

Cultural ecosystem services
Deep learning
River ecosystems
Environmental management
Social media

ABSTRACT

Cultural Ecosystem Services (CES) are essential for human well-being, particularly those provided by river landscapes. Yet, CES remains overlooked in river conservation strategies due to its intangible nature and the methodological challenges involved in their assessment. This study introduces a novel AI-based framework that integrates deep learning for image recognition and machine learning for modelling to assess CES across river landscapes at regional scale. ResNet-152 convolutional neural network was fine-tuned to classify 6911 Flickr images into CES categories. The classified photos were then linked to biophysical variables using an XGBoost model, enabling interpretable predictions of biophysical CES drivers across heterogeneous landscapes. Residual analysis of population-based predictions revealed spatial clusters of “added CES value,” highlighting cultural benefits not explained by demographic factors alone.

This integrated approach goes beyond previous CES assessments by combining automated image classification, large-scale spatial mapping of CES, and interpretable modelling of biophysical variables, allowing the cost-effective identification of under-recognized CES hotspots. Findings highlight the value of quotidian urban rivers and protected areas as key CES hotspots. The framework is transferable, reproducible, and openly available, thereby bridging AI methods and conservation planning.

1. Introduction

River landscapes are significant biodiversity hotspots, yet they are among the most threatened environments on Earth (Strayer, 2010). Beyond their ecological significance, rivers also support human well-being through Cultural Ecosystem Services (CES) (Grizzetti et al., 2016). Defined as the non-material benefits people obtain from ecosystems (Millennium Ecosystem Assessment, 2005), CES are inherently context-dependent, emerge from direct human-nature interactions and are closely linked to the ecological quality of landscapes (Fish et al., 2016; Kosanic and Petzold, 2020).

CES are particularly valued in industrialized societies, where they contribute to leisure, aesthetic enjoyment, and cultural identity (Hernández-Morcillo et al., 2013). Thus, a deeper understanding of the CES provided by river landscapes is crucial for the success of conservation efforts (Chan et al., 2012; Milcu et al., 2013; Nie et al., 2022).

However, CES remain understudied and frequently overlooked in conservation strategies due to their intangible nature, their subjective component, and the challenges of quantifying their spatial distribution

(Hermes et al., 2018). Previous efforts to assess CES have often relied on resource-intensive approaches such as participatory mapping (Xu et al., 2020), surveys (Inácio et al., 2022), interviews (Margaryan et al., 2022), or field observations (Nabout et al., 2022), which are challenging to scale and difficult to replicate (Hernández-Morcillo et al., 2013). Consequently, there is a growing demand for scalable and cost-effective methodologies to integrate CES into conservation planning and policy-making (Schirpke et al., 2023).

Over the past decade, crowdsourced data has become a key methodological advancement in CES evaluation, enabling cost-effective assessments (Langemeyer et al., 2023). Georeferenced user-generated content from social media platforms has proven useful for identifying and evaluating CES. Among those platforms, Flickr is the most used for CES assessment due to its accessible API, high volume of geotagged content, and the integration of photographic and textual data (Da Mota and Pickering, 2020).

Although previous research has demonstrated the efficacy of social media data on local scales (Cheng et al., 2019), its greatest potential lies in large-scale assessments, where it can identify the value of CES across

* Corresponding author. Catalan Institute for Water Research (ICRA – CERCA) Carrer Emili Grahit 101, 17003, Girona, Spain.

E-mail address: fcomalada@icra.cat (F. Comalada).

<https://doi.org/10.1016/j.jenvman.2025.127667>

Received 27 May 2025; Received in revised form 14 September 2025; Accepted 13 October 2025

Available online 18 October 2025

0301-4797/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

extensive regions (Ghermandi and Sinclair, 2019). However, the overwhelming volume of data generated at such scales poses a challenge, making the integration of automated classification methods necessary to extract meaningful insights (Tufekci, 2014).

Artificial intelligence (AI) emerges as a valuable tool to manage and analyze large volumes of data, and thus for supporting large-scale CES assessments (Zhang et al., 2022). Although distinct AI-based methodologies have been employed in CES research (Comalada et al., 2025; Richards and Tunçer, 2018), Convolutional Neural Networks (CNNs) has proven particularly effective enabling transfer learning tailored to CES categories, and often outperforming commercial image-labeling APIs. Reported CES accuracies of CNN models range from 0.77 to 0.83, and F1-scores from 0.59 to 0.83 (Cardoso et al., 2022; Lingua et al., 2022; Egarter Vigl et al., 2021; Winder et al., 2022). Although the outcomes are reliable, their generalizability room for improvement improvement to unseen datasets.

Besides classifying CES, understanding which biophysical variables influence their perception and identifying areas with added CES value can support conservation and restoration strategies (Dou et al., 2022; Poe et al., 2014). However, to the best of our knowledge, the studies that contributed to this field have been conducted on a local scale (Arslan and Özücü, 2021; Clemente et al., 2019; Nyelele et al., 2023; Ridding et al., 2018). Therefore, their application for conservation policies is limited.

Building on these advances, we address two research questions: (i) can a ResNet-152 CNN with transfer learning achieve higher classification accuracy in river-related CES compared to previous studies? and (ii) which biophysical variables best explain CES distribution at large-scale?

Thus, in this study, we developed a methodological framework to model river-related CES spatial distribution using social media photos and biophysical variables at large scales. We pre-trained ResNet-152 CNN (He et al., 2015) using a dataset of 6911 geotagged Flickr images capturing freshwater CES. Then, we analyzed the relationship between CES distribution and biophysical variables using the XGBoost machine learning model (Chen and Guestrin, 2016), and by analyzing model residuals, we identified areas with a high concentration of CES that could not be explained by population density alone, identifying areas with added CES value.

The resulting framework was applied across the Iberian Peninsula rivers, providing an unprecedented spatial representation and analysis of CES, and valuable insights for evidence-based environmental planning and management. Rivers were selected due to their critical relevance for environmental management (Tickner et al., 2020).

The methodological framework is available through an open, user-friendly interface, as well as the employed code, the training and validation dataset, and the generated raster layers (Comalada, 2025).

2. Materials and methods

We first developed a framework to map river CES using social media photos and then tested it across the Iberian Peninsula rivers. First, we collected all geotagged Flickr photos taken in Iberian rivers between 2022 and 2024. Then, a CNN was trained with hashtag-based images to classify photos into five CES categories: Aesthetic values, Recreational values, Cultural Heritage Values, Fauna Fascination, and Flora Fascination. Its performance was tested on 1351 independent images from the Ter River basin. After downloading all photos from the case study, CNN classified them into five categories.

Finally, we trained XGBoost models for each CES category using five environmental predictors (naturalness, accessibility, river order, Natura 2000 and population density), and analyzed spatial patterns using Moran's I, Mann-Whitney *U* test, and AUC statistics.

2.1. Study area

All river landscapes of the Iberian Peninsula were used as case study.

The Iberian Peninsula is in the southwest of Europe, has a surface area of 583,000 km², and It is composed of three states, which are Spain (505,990 km²), Portugal (92,090 km²), and Andorra (468 km²) (Fig. 1).

Its complex orography leads to significant climatic and landscape variability at both local and regional scales (Beck et al., 2023), leading to a highly diverse fluvial network. The Iberian Peninsula also encompasses multiple languages, traditions, and land-use practices shaped by long-standing historical and socio-political trajectories. This diversity, combined with significant human pressure, makes the Iberian rivers one of the most important biodiversity hotspots in the world.

2.2. Data acquisition

First, a 1 km buffer was applied to all rivers within the Iberian Peninsula using the Copernicus river layer (European Environment Agency, 2020). The buffered area was divided in 1 km² cells, and only those covering 40 % or more of the buffer area were selected as extraction zones. The final dataset comprised 184,731 km of length and 312,398 km² of surface in extraction zones (Fig. 1).

Finally, all Flickr geolocated photos published within the extraction zones, and taken between January 1st, 2022, and December 31st, 2024, were downloaded and stored as their Flickr Photo ID in a dedicated folder (flickrapi). To minimize API inconsistencies, the download was performed twice, following Leppämäki et al. (2025).

2.3. CES categorization and labelling

Table 1 details the five selected freshwater-related CES categories used for pre-training the CNN model and for the study case.

Once the categories were defined, a set of Flickr photos was collected to pre-train the model, adding one extra category to discard those photos that were not related to freshwater-CES (Table 1.). To guarantee generalization across diverse river landscapes, the training images were collected from Flickr using keyword-based selection criteria (2022–2024) and without geographical restrictions to avoid spatial bias (Beery et al., 2018).

Before CNN training, three researchers independently classified 100 randomly selected photos to validate the CES classification criteria. Inter-rater reliability showed a high agreement (mean Cohen's kappa = 0.95) (Cohen, 1960) (supplementary material). Then, one researcher manually labeled representative photos for each category (Table 1).

2.4. CNN model development

2.4.1. Architecture and preprocessing

ResNet-152 (He et al., 2015) with pre-trained ImageNet-1K V2 weights (Deng et al., 2009) was trained for CES classification. This architecture was selected for its high performance in image recognition tasks and its proven suitability for transfer learning, particularly in environmental image classification contexts, following evidence from Cardoso et al. (2022), who demonstrated that this configuration yields optimal performance in CES classification tasks.

To optimize training, the RangerAdaBelief optimizer was used (Zhuang et al., 2020). This optimizer combines the benefits of Rectified Adam (RADam) and Lookahead mechanisms and dynamically adapts learning rates based on the "belief" in gradient directions, making it particularly effective for transfer learning scenarios. Early stopping was implemented with a patience threshold of 10 epochs to minimize overfitting by halting training once the validation loss failed to improve consistently.

The loss function used was CrossEntropyLoss, the standard criterion for multi-class classification tasks, as it penalizes the divergence between predicted class probabilities and ground truth labels. All model development and training were conducted using the PyTorch framework (Paszke et al., 2019), within a Python 3.10.3 virtual environment configured for compatibility with PyTorch, TensorFlow (Abadi et al.,

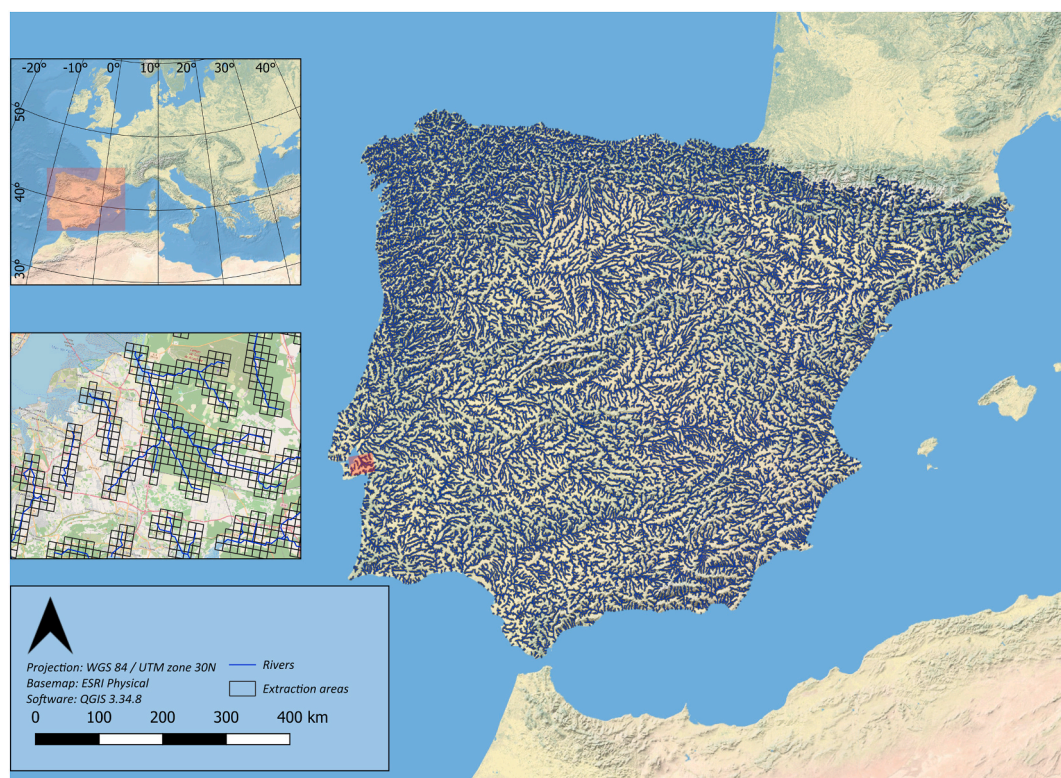


Fig. 1. Location of the Iberian Peninsula, showing the river network and the extraction zones used in the analysis (grid overlay).

Table 1

CES categories used to train the CNN model, including category definitions, keywords used to extract relevant Flickr photos, and the number of images per included in the training dataset. The "Non-river CES" category stands for images unrelated to river CES and was included to help the model distinguish irrelevant photos. Definitions were adapted from [Cheng et al. \(2019\)](#) and [Havinga et al. \(2023\)](#).

Category	Description	Keywords	n
Aesthetic Values	Beauty or spectacularity of natural aspects of freshwater landscapes.	Brooks, Cascade, Creek, FreshwaterBeauty, Gorge, Lagoon, Lake, Pond, Rapids, River, Rivers, Rivulet, Stream, Waterfall, Wetland	1433
Recreational Values	Sportive, leisure or relaxation activities practiced on freshwater landscapes.	Boating, Canyoning, Fishing, Kayak, Kayaking, Lake swimming, Rafting, River recreation, River swimming, Riverboarding	931
Cultural Heritage Values	Appreciation of modern and historical freshwater infrastructures for their functionality, aesthetic appeal, or cultural significance.	Aqueduct, Boat, Bridge, Dam, Drawbridge, Dyke, Hydroelectric, River ferry, Riverboat, Steamboat, Watermill, Weir	608
Fauna Fascination	Beauty and diversity of fauna within freshwater ecosystems.	Amphibian, Animal, Ants, Biodiversity, Bird, Birds, Butterfly, Fauna, Frog, Insect, Lizard, Mammal, Reptile, Wetland fauna, Wildlife	1282
Flora Fascination	Beauty and diversity of flora within freshwater ecosystems.	Angiospermae, Floral, Flora, Flower, Flowers, Flowering plants, Freshwater flora, River flora, River flower, Wildflowers	870
Non-river CES Related	Photographs that do not relate to any river CES.	–	1787

2016), and auxiliary libraries including RangerAdaBelief.

The model was trained using a batch size of 32 and a fixed initial learning rate of 0.001. The CosineAnnealingLR scheduler was configured with $T_{max} = 10$ and $\eta_{min} = 1e-6$. Mixup augmentation was applied with an alpha parameter of 0.4. All random number generators (NumPy, PyTorch) were seeded with a fixed value of 42 to guarantee reproducibility across runs.

2.4.2. Training and validation

All images used for training were resized to 224×224 pixels and normalized for brightness, contrast, and color. The training dataset comprised 6911 labeled Flickr images distributed across six CES categories (Table 1).

To improve generalization, data augmentation techniques were applied, including Mixup ([Zhang et al., 2018](#)), random horizontal flipping, and random rotation ([Shorten and Khoshgoftaar, 2019](#)). For transfer learning, the first two layers of the CNN based on ResNet-152 were frozen, while deeper layers were fine-tuned following [Cardoso et al. \(2022\)](#). The final classification layer was replaced with a custom six-class output and regularized using dropout ([Srivastava et al., 2014](#); [Glorot and Bengio, 2010](#)).

The model was trained using CrossEntropyLoss ([Prechelt, 1998](#)) and optimized with RangerAdaBelief ([Zhuang et al., 2020](#)), applying early stopping with a patience of 10 epochs to prevent overfitting. A 5-fold cross-validation scheme was used, with 80 % of the data for training and 20 % for validation in each fold ([Paszke et al., 2019](#)).

Finally, model performance was validated on an independent test set of 1351 unseen images taken between 2022 and 2024 and published within the Ter River basin (northeastern Spain) ([He et al., 2015](#)).

2.5. Photo classification and metadata filtering

CNN was used to classify all geotagged Flickr photos collected within the extraction zones across the Iberian Peninsula into one of the six predefined CES categories.

For each photo classified as CES, metadata was retrieved using its unique Photo ID through two separate Flickr API calls: one retrieving public information (geographic coordinates and upload dates), and another retrieving private information (user ID).

To reduce overrepresentation, a Photo-User-Day (PUD) filter was applied (Wood et al., 2013). Only one photo per user per day was randomly retained, and all other photos matching the same criteria were discarded (Egarter Vigl et al., 2021).

This produced a spatially balanced dataset of CES-labeled photos with associated geographic coordinates, timestamps, and user identifiers, suitable for subsequent spatial modelling.

2.6. Biophysical model for CES prediction

To model the spatial distribution of CES photos, we applied the Extreme Gradient Boosting (XGBoost) algorithm (Chen and Guestrin, 2016) known for its high accuracy and scalability in heterogeneous spatial data contexts (Elith and Leathwick, 2009). The model was trained separately for each CES category and for the aggregated dataset.

Hyperparameter optimization was carried out using grid search combined with 5-fold cross-validation. The search space included max_depth (3–10), learning_rate (0.01–0.3), n_estimators (100–1000), colsample_bytree (0.5–1.0), and subsample (0.5–1.0). For each candidate configuration, performance was evaluated by minimizing log loss and maximizing AUC across folds. The final selected configuration was max_depth = 6, learning_rate = 0.1, n_estimators = 500, colsample_bytree = 0.8, subsample = 0.8.

Model interpretability was assessed using variable importance scores, which were computed consistently across folds. In addition, permutation importance with 50 repetitions was applied to quantify the effect of each predictor on model performance, and empirical p-values were derived to test whether mean AUC values were significantly greater than zero. SHAP values were then computed to capture both the magnitude and the directionality of each predictor's contribution, with Pearson correlations used to formally test whether predictor values systematically influenced the sign of SHAP effects.

Presence-absence data were constructed using a 1:2 sampling ratio (He and Garcia, 2009), pairing each presence point (location with ≥ 1 CES photo) with two randomly generated absence points within the same extraction area (Barbet-Massin et al., 2012). Each point was linked to five biophysical variables.

1. Naturalness, based on the Hemeroby index derived from CORINE Land Cover 2018; European Environment Agency (2020), Walz and Stein (2014).
2. Accessibility, measured as $\log_{10}$ -transformed Euclidean distance to the nearest major road (Meijer et al., 2018).
3. River Order, extracted from EU-Hydro and assigned the Strahler order of the closest stream segment (Strahler, 1952).
4. Natura 2000, encoded as a binary indicator of protection status (European Environment Agency and European Commission, 2024).
5. Population Density, obtained from WorldPop as a 100 m resolution raster and $\log_{10}$ -transformed (Tatem, 2017).

Predictor selection was based on data availability and relevance for river landscape management. No variables were excluded due to low multicollinearity (Pearson's $r < 0.3$; Dormann et al., 2013). The model was trained with 5-fold cross-validation (80 % training, 20 % validation).

To analyze the spatial effect of Population Density, a XGBoost model was trained using only this variable. The residuals were interpolated using Inverse Distance Weighting (IDW) and mapped as 1 km resolution raster layers. Spatial autocorrelation in residuals was evaluated using Moran's I with a Queen contiguity matrix (Anselin, 1995). Final raster maps are available in GitHub repository (Comalada, 2025).

To examine the influence of Natura 2000 in zones with added CES

value, Population Density residual values were extracted for pixels located inside and outside the Natura 2000 network, based on spatial intersection with protected area boundaries. A two-sided Mann–Whitney U test was then applied to assess whether the distribution of residuals differed significantly between protected and unprotected areas. The U statistics were then normalized to compute the Area Under the Curve (AUC), providing an interpretable, non-parametric effect size metric ranging from 0 to 1.

3. Results

3.1. AI model performance

Table 2 summarizes the performance of the ResNet-152 model, which achieved an overall accuracy of 0.91 on an independent test set composed of unseen images posted in the Ter River Basin.

The Aesthetic Values category showed strong performance (F1 = 0.89), with high recall (0.93) indicating effective identification of true positives. Slightly lower precision (0.86) suggests some false positives, likely due to visual overlaps with natural elements like vegetation or scenic views unrelated to rivers.

Recreational Values achieved high precision (0.92), but lower recall (0.74), indicating a conservative classification: the model avoided false positives but missed some true instances. Similarly, Cultural Heritage Values also had high overall performance (F1 = 0.91) but reduced recall (0.85), likely due to visual similarity with aesthetic or non-river-related historical scenes.

In contrast, Fauna Fascination and Flora Fascination exhibited high performance, with accuracy and F1-scores of 0.94 and 0.93, and 0.99 and 0.97, respectively.

For the Non-River CES Related, recall was high (0.96), suggesting effective identification of irrelevant content, although precision (0.87) indicates occasional misclassification of CES photos.

3.2. Spatial CES distribution

A total of 754,676 photographs were classified by the model. Of these, 129,968 were classified as river CES (17 %). After applying PUD filter, 40,743 photos were selected (5 %) (Fig. 2).

Aesthetic Values (Fig. 2a) and Cultural Heritage Values (Fig. 2c) were the most popular, both reflecting visually appealing landscapes. Fauna Fascination (Fig. 2d) was mostly associated with unpopulated areas, while Flora Fascination (Fig. 2e) was more evenly distributed throughout the region. Recreational Values (Fig. 2b) were the least popular category in this analysis. These were distributed along the Duero River and river mouths, where they often overlap with marine influences. Temporal distribution of the analyzed publications is available at Annex 1 (supplementary materials). This shows small interannual differences in the total number of publications and their distribution between CES categories.

3.3. CES modelling

The XGBoost model demonstrated strong predictive performance in

Table 2
Resnet-152 model performance metrics results by category.

Category	Precision	Recall	f1-score	Support
Aesthetic Values	0.86	0.93	0.89	248
Recreational Values	0.92	0.74	0.82	31
Cultural Heritage Values	0.98	0.85	0.91	363
Fauna Fascination	0.94	0.92	0.93	91
Flora Fascination	0.99	0.95	0.97	214
Non-River CES Related	0.87	0.96	0.91	404
Macro Average	0.93	0.89	0.91	1351
Weighted Average	0.92	0.91	0.92	1351

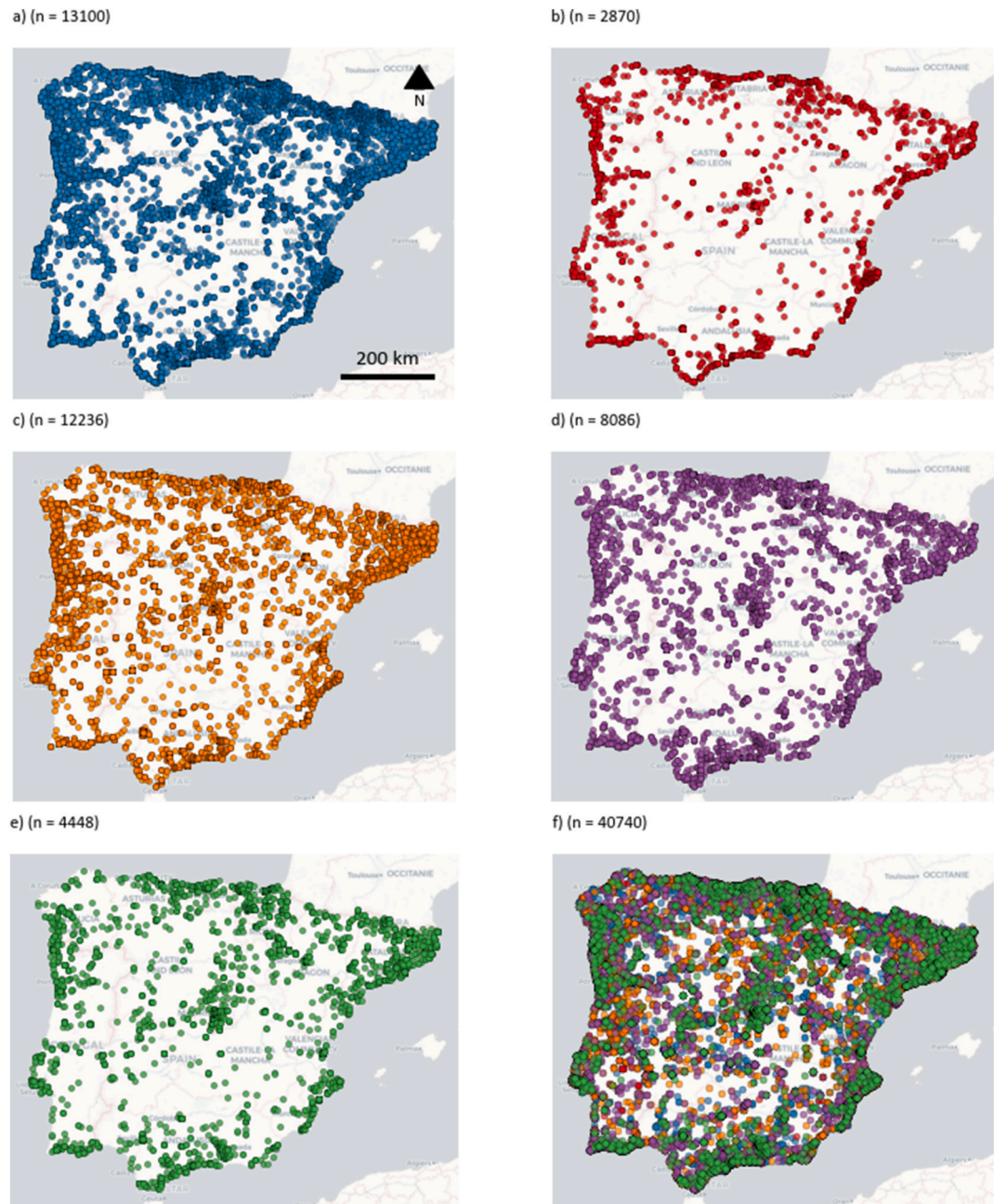


Fig. 2. Spatial distribution of PUD per CES category. a) Aesthetic Values, b) Recreational Values, c) Cultural Heritage Values, d) Fauna Fascination, e) Flora Fascination, f) All CES Combined.

identifying biophysical variables associated with the presence of all CES, and for each CES category (Table 3).

The contribution of predictor variables varied across CES categories (Fig. 3). Population Density was consistently the most influential predictor, producing the largest performance gains when included (AUC =

0.21–0.32 across models). Natura 2000, Naturalness, Accessibility, and River Order contributed smaller but statistically significant effects (AUC <0.06). These values indicate that while demographic pressure dominates the predictive signal, ecological and management-related factors provide additional explanatory power that systematically improves model accuracy. SHAP results further demonstrated that the influence of Natura 2000 and Naturalness was consistently positive, whereas Accessibility and River Order tended to reduce CES probability.

Model accuracy varied across categories, ranging from 0.80 in Aesthetic values (Fig. 3a) to 0.90 in Cultural Heritage Values (Fig. 3c). Population Density was consistently the top predictor, explaining 24 %–43 % of the variance, except in Aesthetic Values, where Natura 2000 contributed most (33 %).

Natura 2000 influence was substantial across models (12 %–31 %). Naturalness (15 %–20 %), Accessibility (14 %–17 %), and River Order

Table 3
Summary of XGBoost model performance metrics for each CES category.

Value Type	Accuracy	F1-score	Macro F1	Weighted F1
Aesthetic values	0.80	0.68	0.77	0.80
Recreational values	0.83	0.73	0.80	0.82
Cultural Heritage values	0.90	0.85	0.89	0.90
Fauna Fascination	0.85	0.77	0.83	0.85
Flora Fascination	0.84	0.76	0.82	0.84
All CES combined	0.85	0.76	0.82	0.84



Fig. 3. Variable importance for the classification models trained to predict the presence of different CES. a) Aesthetic Values, b) Recreational Values, c) Cultural Heritage Values, d) Fauna Fascination, e) Flora Fascination, f) All CES Combined.

(12 %–14 %) showed lower but more stable contributions.

3.4. Spatial analysis of residuals

Residual analysis of the population density revealed consistent spatial autocorrelation across all CES categories, identifying areas with added CES value (Fig. 4). Moran's I values ranged from 0.25 to 0.35 for individual categories ($p < 0.001$), with the highest spatial structure observed in Fig. 4f (All CES Combined, Moran's I = 0.47, $p < 0.001$). This suggests that additional biophysical or cultural variables significantly contribute to the spatial configuration of CES beyond population distribution.

Residuals were significantly less negative inside Natura 2000 areas across all CES categories ($p < 0.001$), indicating consistently higher added value within protected sites. AUC values ranged from 0.56 to 0.62, with the strongest contrast in the combined CES layer (AUC = 0.62), followed by Aesthetic Values (0.61) and Fauna Fascination (0.60).

4. Discussion

This study contributes to answering diverse knowledge gaps in CES research. By identifying and modelling CES across river landscapes at a large spatial scale, we respond to the need for policy-relevant indicators (Oteros-Rozas et al., 2018) and scalable frameworks that support the integration of CES into ES modelling efforts (Schirpke et al., 2023). Furthermore, this study models the biophysical variables that influence CES in river landscapes, one of the most neglected landscapes in CES research (Kosanic and Petzold, 2020; Thiele et al., 2019), although one of the most threatened and valuable (Sayer et al., 2025). Recent European policy agendas, such as the EU Biodiversity Strategy for 2020 (EU Biodiversity Strategy for 2020, 2011), have increasingly emphasized the integration of CES into environmental planning, promoting the use of scalable, data-driven approaches (European Environment Agency, 2020;

European Commission, 2016). This reflects a broader shift from traditional assessments to automatic and cost-effective ones (Plieninger et al., 2013). In addition, by assessing CES across a diverse set of river landscapes, we highlight the relevant role that protected areas play in sustaining CES in comparison to other spaces, thereby recognizing their non-material contributions to human well-being (Maes et al., 2012). Thus, the framework is available in a user-friendly Python notebook (Comalada et al., 2025), to facilitate replication and transparency (Schirpke et al., 2023).

4.1. Methodological considerations and limitations

A key limitation of this study is the representativeness of the data source. Flickr, while valuable for spatially explicit research (Da Mota and Pickering, 2020) does not reflect the broader population's demographics and online behavior (Leppämäki et al., 2025). This introduces systematic biases: Flickr users are typically younger, urban-based, and more digitally active, which means that the results primarily capture the preferences of this subset of society rather than those of the population at large. In addition, spatial posting patterns are inherently uneven, with photographs clustering around visually appealing or touristic sites, leading to an overrepresentation of certain river stretches and underrepresentation of more ordinary or less photogenic landscapes (Do et al., 2015; Llanos-Paez and Acuna, 2022). These demographic and spatial biases should therefore be considered when interpreting CES patterns, as they influence both the intensity and distribution of the mapped services. Nevertheless, several studies demonstrate that Flickr activity correlates with actual visitation rates to natural areas, supporting its use as a proxy for recreational and aesthetic preferences (Hausmann et al., 2018; Sinclair et al., 2020; Tenkanen et al., 2017).

Technical limitations also affect data quality and reproducibility. The Flickr API can exhibit inconsistencies in data retrieval, posing

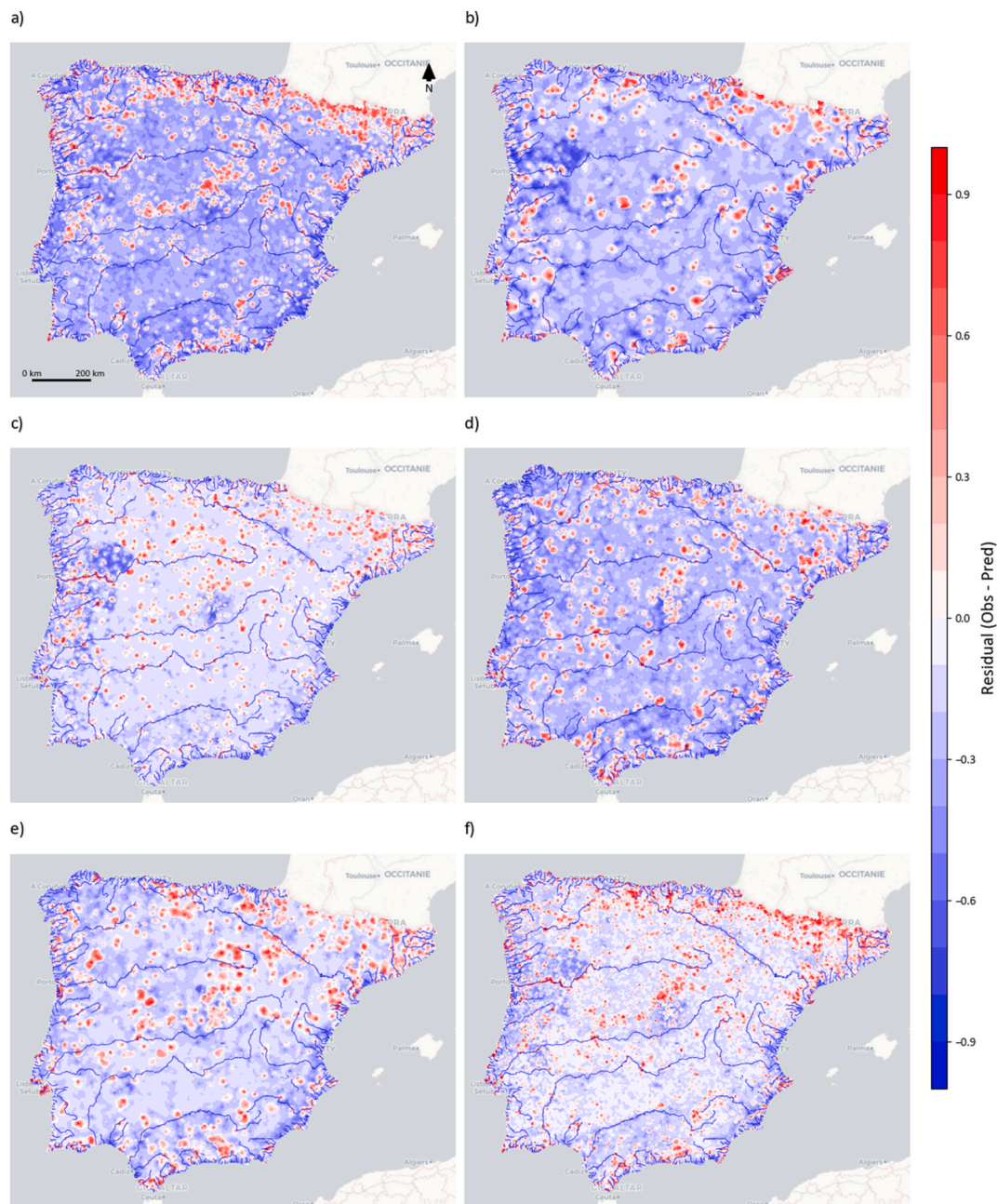


Fig. 4. Spatial distribution of model residuals for each CES category. These spaces are referred as areas with added CES value. a) Aesthetic Values, b) Recreational Values, c) Cultural Heritage Values, d) Fauna Fascination, e) Flora Fascination, f) All CES Combined.

challenges for study replication and long-term monitoring efforts (Leppämäki et al., 2025). Additionally, Zielstra & Hochmair (2013) demonstrated that manually geotagged photos often introduce spatial inaccuracies, leading to significant spatial mismatches.

Another limitation lies in its dependence on visual data. Only CES dimensions that can be captured through photographs are represented, while many important CES such as spiritual values, sense of place, or symbolic meanings cannot be visualized and thus remain unassessed in this framework (Chan et al., 2012). Furthermore, the model assigns each image to a single CES category, which may oversimplify reality, especially in complex landscapes (Fish et al., 2016). Additionally, PUD filter randomly selects just one photo, potentially excluding CES images. This categorical simplification also reflects the inherent ambiguity in CES boundaries, as many images could reasonably belong to multiple categories simultaneously.

The selection of biophysical variables is another limitation. Our model focused on variables (naturalness, accessibility, Natura 2000, ecological status and river order) that are directly relevant to river landscape management, and can be influenced through planning or policy interventions. We did not include factors such as climate or soil type since these cannot be modified by management actions and would therefore contribute less directly to actionable conservation strategies.

Finally, ethical considerations must be acknowledged. Although all data used in this study were obtained through Flickr's public API, and only publicly available photographs were included, the metadata collected included sensitive information, as geolocation, user id, or publication timestamps. To mitigate this, results were interpreted without reference to individual users, considering the ethical responsibility that comes with working with user-generated content.

4.2. AI classification model

CNN emerged as a promising resource for evaluating CES at large-scale. However, only four studies have validated their CNN models using unseen images. Egarter Vigl et al. (2021) (Clarifai General Model v1.3) achieved 0.82 accuracy and an F1 score of 0.80. Cardoso et al. (2022) (ResNet-152) reported a global classification accuracy of 0.77 and an F1 score of 0.69. Lingua et al. (2022) (ResNet-152) reported 0.80 accuracy and an F1 score of 0.83, and Winder et al. (2022) (Inception-ResNetV2) achieved 0.71 accuracy and an F1 score of 0.61. These suggest a common limitation in the generalizability of CNN models when applied to unseen datasets. In contrast, our model (ResNet-152), which predicted similar categories, achieved a classification accuracy exceeding 0.91 on an unseen dataset, with an F1 score of 0.91, accomplishing a substantial improvement in predictive performance. Importantly, performance varied notably across CES categories (F1 ranging from 0.82 to 0.97). Lower scores in Recreational, Aesthetic, and Cultural Heritage values were linked to their higher visual heterogeneity and overlap with non-river elements, whereas Fauna and Flora Fascination achieved higher scores due to their distinctive visual patterns.

The model improvement can be attributed to four factors. First, our training dataset was compiled using images extracted using thematic tags, whereas most previous studies extracted images from a single geographic region (Cardoso et al., 2022; Havinga et al., 2024). This approach resulted in a highly diverse and representative dataset. Classes were balanced in both volume and thematic clarity, and no spatial sampling bias was introduced. This improved the model's capacity to generalize across diverse geographic contexts, as required for large-scale assessments (Beery et al., 2018). Furthermore, the tag-based extraction method introduced substantial landscape variability, enabling applicability to multiple regions worldwide and facilitating the integration of social media images into large-scale river CES assessments.

Second, all images used in training and validation were captured between 2022 and 2024, a period when most smartphones were equipped with high-resolution cameras, establishing consistency in image quality. This is significant considering that ResNet-152 resizes input images to 224×224 pixels (Dodge and Karam, 2016).

Third, our model employs RangerAdaBelief optimizer (Zhuang et al., 2020), an advanced algorithm that combines adaptive learning rate strategies with mechanisms designed to improve generalization performance. Its capacity to avoid overfitting contributed to the strong performance observed with unseen data.

Fourth, our model was trained specifically to identify features within aquatic landscapes. These possess unique spectral and textural characteristics including smoothness, reflectivity, and color consistency (Z. Zhang et al., 2021), which likely improved the model's classification sensitivity for photos linked to freshwater landscapes.

4.3. Biophysical CES model

The XGBoost model demonstrated strong performance in identifying the biophysical variables that influence CES presence (0.85 accuracy, 0.84 weighted F1 score), offering a reliable and interpretable approach: The model architecture is simple and transparent, and its outputs are intuitive and easy to interpret, enabling the participation of decision makers when it comes to replicating the methodology or analyzing results. Moreover, the model allows researchers to consider social generated data, differing from most of the previous models, which are trained often on biophysical variables (Christie et al., 2019).

As expected, population density emerged as the strongest predictor in the spatial models, reflecting the baseline intensity of photographic activity rather than the absolute provision of CES. We therefore interpret population density not as a direct biophysical driver, but as a proxy for the likelihood of social media presence. To overcome this limitation, we used residual analysis, which identifies locations where CES values are higher than predicted solely by demographic pressure. Residuals are not

random artefacts but systematically align with ecologically and socially significant contexts, such as Natura 2000 protected areas and urban rivers. Thus, residuals represent an “added CES value,” capturing CES associated with ecological quality and everyday interactions that extend beyond population-driven photographic activity.

At the category level, this predominance of population density underscores the role of urban rivers in delivering CES. These include the ornamental value of flowers in river landscapes (Oteros-Rozas et al., 2018), everyday wildlife encounters (Cox and Gaston, 2016), the cultural significance of river-related infrastructure (Plieninger et al., 2013), and the recreational opportunities rivers provide in densely populated areas (Grzyb, 2024). However, Aesthetic Values, the most popular CES category, also shows strong associations with Natura 2000, suggesting that people are willing to travel to visit landscapes with high ecological status, perceived as visually appealing (Hausmann et al., 2018). This interpretation is reinforced by the Naturalness variable, which explains nearly 20 % of the variance.

This is supported by the residuals derived from models using only Population Density as a predictor, which allowed the identification of areas with added CES value. These residuals provide valuable insights for spatial planning and conservation, as they highlight locations where CES exceed expectations based solely on demographic pressure.

Across all CES categories, residual values were consistently higher within Natura 2000 sites. The strongest associations were observed for the combined CES model (AUC = 0.62), Aesthetic Values (AUC = 0.61), and Fauna Fascination (AUC = 0.60), indicating a marked preference for visually and biologically rich landscapes. Flora Fascination, Cultural Heritage Values, and Recreational Values showed positive, albeit weaker associations (AUC = 0.57, 0.56, and 0.56, respectively). These results suggest that such services are more frequently accessed in everyday environments rather than through targeted travel.

4.4. Management implications

This study demonstrates that analyzing the relationships between biophysical variables and river-related CES distribution provides valuable insights into spatial planning, thereby supporting landscape conservation and management. Results suggest that CES provision is driven by a combination of biophysical variables, with Population Density and the presence of Natura 2000 network emerging as the most influential. Although all river landscapes have the potential to provide CES, the predominance of urban rivers in our results should be interpreted with caution, as it may partly reflect photographic convenience rather than CES provision. In this sense, our findings indicate patterns of perceived cultural benefits rather than definitive measures of CES supply. Nonetheless, urban rivers remain highly relevant because they significantly improve public access to CES and contribute to human health and well-being (Kaiser et al., 2021; Pouso et al., 2018), reinforcing previous evidence that restoration efforts should not be limited to pristine or unmodified rivers but also target everyday urban waterways (Garcia et al., 2020; Völker and Kistemann, 2011).

Natura 2000 also explains a significant share of CES presence, suggesting that river landscapes in protected areas are not only valued for their biodiversity, but also for their beauty and leisure potential. This added CES value reinforces the role of protected areas as spaces that strengthen human physical and psychological wellbeing (Terraube et al., 2017), while increasing public support and social acceptance (Whitelaw et al., 2014). Furthermore, nature-based tourism provides revenues to different stakeholders and facilitates consensus about the conservation of those valued spaces (Di Minin et al., 2013). However, we also acknowledge that the popularity of protected areas and wilderness zones can generate negative ecological impacts through habitat disturbance, pollution, and ecological degradation (Marion et al., 2016). Social media can amplify these pressures by promoting specific destinations and homogenizing visitation patterns, which concentrates visitor flows and increases local ecological stress (Becken et al., 2017).

For this reason, our emphasis on protected areas is now framed not only as an opportunity for linking human well-being and conservation goals, but also as a management challenge that requires balancing access with ecological sustainability.

As reported in Section 3.4, the AUC value of all CES model residuals (0.62) suggests that areas with added CES value like La Vall Fosca (located near an IUCN Category II National Park (Dudley, 2008) in northwestern Spain) often lack formal protection, suggesting that several sparsely populated areas experiencing intense human pressure remain unregulated.

By modelling CES presence as a function of population density and analyzing residuals, it is possible to identify such pressured natural river landscapes at a broad spatial scale, thereby supporting efforts to redefine protection priorities. Furthermore, the generated cartography can be integrated with other regional ES models to estimate the value of the ecosystems of a given territory (Schirpke et al., 2023).

Redefining policies based on the identification of crowdsourced CES hotspots can improve the management of ecologically valuable river landscapes that currently lack formal protection, thereby contributing to more effective nature conservation. In parallel, restoring the ecological quality of urban river landscapes could serve as a strategic approach to redistribute visitor flows, mitigate anthropogenic pressure on protected areas, and increase CES delivery in densely populated areas (Brückner et al., 2021; Garcia et al., 2017).

5. Conclusions

The present study introduces and tests a scalable framework that combines deep learning-based image classification with biophysical modelling to assess CES across river landscapes. By fine-tuning a ResNet-152 convolutional neural network and integrating it with a XGBoost model trained on geotagged Flickr images, we achieved high content classification accuracy (91 %) and strong biophysical modelling predictive performance (85 %). In addition, the residuals from the CES–population density model enabled the detection of areas with high CES value not explained by population density, providing valuable insight for spatial planning.

Our results confirm that CES provision in river landscapes is shaped by a combination of biophysical variables. While urban rivers emerge as key providers of CES, Natura 2000 sites continue to support high CES values associated with aesthetics and biodiversity observation. Moreover, by modelling XGBoost residuals, areas with added CES value in low-populated spaces were identified.

This research addresses critical gaps in CES research by demonstrating the applicability of artificial intelligence to large-scale landscape assessment. It also reinforces the relevance of river landscapes and emphasizes the role of protected areas and urban systems in supporting non-material human–nature interactions. By identifying under-recognized CES hotspots through crowdsourced data, this approach provides spatially explicit evidence to guide the redefinition of conservation priorities. Integrating this information into environmental governance can contribute to more efficient conservation planning.

CRedit authorship contribution statement

Francesc Comalada: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vicenç Acuña:** Supervision, Resources, Project administration, Conceptualization. **Xavier Garcia:** Writing – review & editing, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Chat GPT4.5 in order to assist in checking the article grammatically and

orthographically, as the authors are not native English speakers. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the European Union Horizon 2020 project MERLIN (H2020-LC-GD-2020-3: 101036337). Authors acknowledge the support from the Economy and Knowledge Department of the Catalan Government through Consolidated Research Groups (ICRA-ENV 2021 SGR 01282), as well as from the CERCA program. F. Comalada acknowledges funding from the Department of Research and Universities of the Generalitat de Catalunya and the European Social Fund for her FI fellowship (2024 FI-1 00133). Open Access funding provided thanks to the CRUE-CSIC agreement with Elsevier.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.127667>.

Data availability

<https://github.com/francesc30/Cultural-Ecosystem-Modelling>

References

- Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., et al., 2016. TensorFlow: a system for large-scale machine learning. arXiv preprint arXiv:1605.08695. <https://doi.org/10.48550/arXiv.1605.08695>.
- Anselin, L., 1995. *Spatial Econometrics: Methods and Models*. Springer Science & Business Media.
- Arslan, E.S., Öricü, Ö.K., 2021. MaxEnt modelling of the potential distribution areas of cultural ecosystem services using social media data and GIS. *Environ. Dev. Sustain.* 23 (2), 2655–2667. <https://doi.org/10.1007/s10668-020-00692-3>.
- Barbet-Massin, M., Jiguet, F., Albert, C.H., Thuiller, W., 2012. Selecting pseudo-absences for species distribution models: how, where and how many? *Methods Ecol. Evol.* 3 (2), 327–338. <https://doi.org/10.1111/j.2041-210X.2011.00172.x>.
- Beck, H.E., McVicar, T.R., Vergopolan, N., Berg, A., Lutsko, N.J., Dufour, A., Zeng, Z., Jiang, X., van Dijk, A.I.J.M., Miralles, D.G., 2023. High-resolution (1 km) köppen-geiger maps for 1901–2099 based on constrained CMIP6 projections. *Sci. Data* 10 (1), 724. <https://doi.org/10.1038/s41597-023-02549-6>.
- Becken, S., Stantic, B., Chen, J., Alaei, A.R., Connolly, R.M., 2017. Monitoring the environment and human sentiment on the great barrier reef: assessing the potential of collective sensing. *J. Environ. Manag.* 203, 87–97. <https://doi.org/10.1016/j.jenvman.2016.06.035>.
- Beery, S., Horn, G., van, Perona, P., 2018. Recognition in Terra incognita (No. arXiv: 1807.04975). arXiv. <https://doi.org/10.48550/arXiv.1807.04975>.
- Brückner, A., Falkenberg, T., Heinzel, C., Kistemann, T., 2021. The regeneration of urban blue spaces: a public health intervention? Reviewing the evidence. *Front. Public Health* 9, 782101. <https://doi.org/10.3389/fpubh.2021.782101>.
- Cardoso, A.S., Renna, F., Moreno-Llorca, R., Alcaraz-Segura, D., Tabik, S., Ladle, R.J., Vaz, A.S., 2022. Classifying the content of social media images to support cultural ecosystem service assessments using deep learning models. *Ecosyst. Serv.* 54, 101410. <https://doi.org/10.1016/j.ecoser.2022.101410>.
- Chan, K.M.A., Satterfield, T., Goldstein, J., 2012. Rethinking ecosystem services to better address and navigate cultural values. *Ecol. Econ.* 74, 8–18. <https://doi.org/10.1016/j.ecolecon.2011.11.011>.
- Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Cheng, X., Van Damme, S., Li, L., Uytendhoeve, P., 2019. Evaluation of cultural ecosystem services: a review of methods. *Ecosyst. Serv.* 37, 100925. <https://doi.org/10.1016/j.ecoser.2019.100925>.
- Christie, M., Martín-López, B., Church, A., Siwicki, E., Szymonczyk, P., Mena Sauterel, J., 2019. Understanding the diversity of values of “Nature’s contributions to people”: insights from the IPBES assessment of Europe and central Asia. *Sustain. Sci.* 14 (5), 1267–1282. <https://doi.org/10.1007/s11625-019-00716-6>.
- Clemente, P., Calvache, M., Antunes, P., Santos, R., Cerdeira, J.O., Martins, M.J., 2019. Combining social media photographs and species distribution models to map

- cultural ecosystem services: the case of a natural park in Portugal. *Ecol. Indic.* 96, 59–68. <https://doi.org/10.1016/j.ecolind.2018.08.043>.
- Cohen, J., 1960. A coefficient of agreement for nominal scales. *Educ. Psychol. Meas.* 20 (1), 37–46. <https://doi.org/10.1177/001316446002000104>.
- Comalada, F., 2025. Cultural ecosystem services modelling [GitHub]. <https://github.com/francesc30/Cultural-Ecosystem-Services-Modelling>.
- Comalada, F., Llorente, O., Acuña, V., Saló, J., García, X., 2025. Using georeferenced text from social media to map the cultural ecosystem services of freshwater ecosystems. *Ecosyst. Serv.* 72, 101702. <https://doi.org/10.1016/j.ecoser.2025.101702>.
- Cox, D.T.C., Gaston, K.J., 2016. Urban bird feeding: connecting people with nature. *PLoS One* 11 (7), e0158717. <https://doi.org/10.1371/journal.pone.0158717>.
- Da Mota, V.T., Pickering, C., 2020. Using social media to assess nature-based tourism: current research and future trends. *Journal of Outdoor Recreation and Tourism* 30, 100295.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L., 2009. ImageNet: a large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition. IEEE, pp. 248–255. <https://doi.org/10.1109/CVPR.2009.5206848>.
- Di Minin, E., Macmillan, D.C., Goodman, P.S., Escott, B., Slotow, R., Moilanen, A., 2013. Conservation businesses and conservation planning in a biological diversity hotspot. *Conserv. Biol.: The Journal of the Society for Conservation Biology* 27 (4), 808–820. <https://doi.org/10.1111/cobi.12048>.
- Do, Y., Kim, J.Y., Lineman, M., Kim, D.-K., Joo, G.-J., 2015. Using internet search behavior to assess public awareness of protected wetlands. *Conserv. Biol.: The Journal of the Society for Conservation Biology* 29 (1), 271–279. <https://doi.org/10.1111/cobi.12419>.
- Dodge, S., Karam, L., 2016. Understanding how image quality affects deep neural networks (No. arXiv:1604.04004). arXiv. <https://doi.org/10.48550/arXiv.1604.04004>.
- Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J.R.G., Gruber, B., Lafrourcade, B., Leitão, P.J., Münkemüller, T., McClean, C., Osborne, P.E., Reineking, B., Schröder, B., Skidmore, A.K., Zurell, D., Lautenbach, S., 2013. Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography* 36 (1), 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>.
- Dou, Y., Zhen, L., Bakker, M., Yu, X., Carsjens, G.J., Liu, J., De Groot, R., 2022. Investigating the potential impact of ecological restoration strategies on people-landscape interactions through cultural ecosystem services: a case study of xilin gol, China. *J. Environ. Manag.* 316, 115185. <https://doi.org/10.1016/j.jenvman.2022.115185>.
- Dudley, N. (Ed.), 2008. *Guidelines for Applying Protected Area Management Categories*. IUCN, Gland, Switzerland.
- Egarter Vigl, L., Marsoner, T., Giombini, V., Pecher, C., Simion, H., Stemle, E., Tasser, E., Depellegrin, D., 2021. Harnessing artificial intelligence technology and social media data to support Cultural Ecosystem service assessments. *People and Nature* 3 (3), 673–685. <https://doi.org/10.1002/pan3.10199>.
- Elith, J., Leathwick, J.R., 2009. Species distribution models: ecological explanation and prediction across space and time. *Annual Review of Ecology, Evolution, and Systematics* 40, 677–697. <https://doi.org/10.1146/annurev.ecolsys.110308.120159>, 40, 2009.
- EU Biodiversity Strategy for 2020, 2011. *Our life insurance, our natural capital: an EU biodiversity strategy to 2020*. European commission. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52011DC0244>.
- European Commission, 2016. Operational potential of ecosystem research applications (OPERAs). <https://cordis.europa.eu/project/id/308393>.
- European Environment Agency, 2020. CORINE Land Cover (CLC) 2018, Version 2020. 20u1 — Copernicus Land Monitoring Service. European Union. <https://land.copernicus.eu/pan-european/corine-land-cover/clc2018>.
- European Environment Agency, 2020. EU-Hydro – River Network Database (Version 1.3). Copernicus Land Monitoring Service. <https://doi.org/10.2909/393359a7-7ebd-4a52-80ac-1a18d5f3db9c> [D at aset].
- European Environment Agency & European Commission, 2024. Natura 2000 dataset. <https://www.eea.europa.eu/data-and-maps/data/natura-11>.
- Fish, R., Church, A., Winter, M., 2016. Conceptualising cultural ecosystem services: a novel framework for research and critical engagement. *Ecosyst. Serv.* 21, 208–217. flickrapi: The python interface to the flickr API (Version 2.4.0). (n.d.). [Python; OS Independent]. Retrieved May 6, 2025, from <https://stuvel.eu/flickrapi>.
- García, X., Benages-Albert, M., Pavón, D., Ribas, A., García-Aymerich, J., Vall-Casas, P., 2017. Public participation GIS for assessing landscape values and improvement preferences in urban stream corridors. *Appl. Geogr.* 87, 184–196. <https://doi.org/10.1016/j.apgeog.2017.08.009>.
- García, X., Benages-Albert, M., Buchecker, Matthias, Vall-Casas, P., 2020. River rehabilitation: preference factors and public participation implications. *J. Environ. Plann. Manag.* 63 (9), 1528–1549. <https://doi.org/10.1080/09640568.2019.1680353>.
- Ghermandi, A., Sinclair, M., 2019. Passive crowdsourcing of social media in environmental research: a systematic map. *Glob. Environ. Change* 55, 36–47. <https://doi.org/10.1016/j.gloenvcha.2019.02.003>.
- Glorot, X., Bengio, Y., 2010. Understanding the difficulty of training deep feedforward neural networks. Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, pp. 249–256. In: <https://proceedings.mlr.press/v9/glorot10a.html>.
- Grizzetti, B., Lanzanova, D., Lique, C., Reynaud, A., Cardoso, A.C., 2016. Assessing water ecosystem services for water resource management. *Environ. Sci. Pol.* 61, 194–203. <https://doi.org/10.1016/j.envsci.2016.04.008>.
- Grzyb, T., 2024. Recreational use of the urban riverscape: what brings people to the river? *Morav. Geogr. Rep.* 32 (1), 14–25. <https://doi.org/10.2478/mgr-2024-0002>.
- Hausmann, A., Toivonen, T., Slotow, R., Tenkanen, H., Moilanen, A., Heikinheimo, V., Di Minin, E., 2018. Social media data can be used to understand tourists' preferences for nature-based experiences in protected areas. *Conserv. Lett.* 11 (1), e12343. <https://doi.org/10.1111/conl.12343>.
- Havinga, I., Marcos, D., Bogaart, P., Massimino, D., Hein, L., Tuia, D., 2023. Social media and deep learning reveal specific cultural preferences for biodiversity. *People and Nature* 5 (3), 981–998. <https://doi.org/10.1002/pan3.10466>.
- Havinga, I., Marcos, D., Bogaart, P., Tuia, D., Hein, L., 2024. Understanding the sentiment associated with cultural ecosystem services using images and text from social media. *Ecosyst. Serv.* 65, 101581. <https://doi.org/10.1016/j.ecoser.2023.101581>.
- He, H., García, E.A., 2009. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* 21 (9), 1263–1284. <https://doi.org/10.1109/TKDE.2008.239>.
- He, K., Zhang, X., Ren, S., Sun, J., 2015. Deep residual learning for image recognition (No. arXiv:1512.03385). arXiv. <https://doi.org/10.48550/arXiv.1512.03385>.
- Hermes, J., Van Berkel, D., Burkhard, B., Plieninger, T., Fagerholm, N., von Haaren, C., Albert, C., 2018. Assessment and valuation of recreational ecosystem services of landscapes. *Ecosyst. Serv.* 31, 289–295. <https://doi.org/10.1016/j.ecoser.2018.04.011>.
- Hernández-Morcillo, M., Plieninger, T., Bieling, C., 2013. An empirical review of cultural ecosystem service indicators. *Ecol. Indic.* 29, 434–444. <https://doi.org/10.1016/j.ecolind.2013.01.013>.
- Inácio, M., Gomes, E., Bogdžević, K., Kalinauskas, M., Zhao, W., Pereira, P., 2022. Mapping and assessing coastal recreation cultural ecosystem services supply, flow, and demand in Lithuania. *J. Environ. Manag.* 323, 116175. <https://doi.org/10.1016/j.jenvman.2022.116175>.
- Kaiser, N.N., Ghermandi, A., Feld, C.K., Hershkovitz, Y., Palt, M., Stoll, S., 2021. Societal benefits of river restoration – implications from social media analysis. *Ecosyst. Serv.* 50, 101317. <https://doi.org/10.1016/j.ecoser.2021.101317>.
- Kosanic, A., Petzold, J., 2020. A systematic review of cultural ecosystem services and human wellbeing. *Ecosyst. Serv.* 45, 101168. <https://doi.org/10.1016/j.ecoser.2020.101168>.
- Langemeyer, J., Ghermandi, A., Keeler, B., van Berkel, D., 2023. The future of crowd-sourced cultural ecosystem services assessments. *Ecosyst. Serv.* 60, 101518. <https://doi.org/10.1016/j.ecoser.2023.101518>.
- Leppämäki, T., Heikinheimo, V., Eklund, J., Hausmann, A., Toivonen, T., 2025. The rise and fall of the social media platform Flickr: implications for nature recreation research. *Journal of Outdoor Recreation and Tourism* 50, 100880. <https://doi.org/10.1016/j.jort.2025.100880>.
- Lingua, F., Coops, N.C., Griess, V.C., 2022. Valuing cultural ecosystem services combining deep learning and benefit transfer approach. *Ecosyst. Serv.* 58, 101487. <https://doi.org/10.1016/j.ecoser.2022.101487>.
- Llanos-Paez, O., Acuna, V., 2022. Analysis of the socio-ecological drivers of the recreational use of temporary streams and rivers. *Sci. Total Environ.* 807, 150805. <https://doi.org/10.1016/j.scitotenv.2021.150805>.
- Maes, J., Egoh, B., Willemen, L., Lique, C., Vihervara, P., Schägner, J.P., Grizzetti, B., Drakou, E.G., Nolte, A.L., Zulian, G., Bouraoui, F., Luisa Paracchini, M., Braat, L., Bidoglio, G., 2012. Mapping ecosystem services for policy support and decision making in the European union. *Ecosyst. Serv.* 1 (1), 31–39. <https://doi.org/10.1016/j.ecoser.2012.06.004>.
- Margaryan, L., Prince, S., Ioannides, D., Roslmaier, M., 2022. Dancing with cranes: a humanistic perspective on cultural ecosystem services of wetlands. *Tour. Geogr.* 24 (4–5), 501–522. <https://doi.org/10.1080/14616688.2018.1522512>.
- Marion, J.L., Leung, Y.-F., Eagleston, H., Burroughs, K., 2016. A review and synthesis of recreation ecology research findings on visitor impacts to wilderness and protected natural areas. *J. For.* 114 (3), 352–362. <https://doi.org/10.5849/jof.15-498>.
- Meijer, J.R., Huijbregts, M.A.J., Schotten, K.C.G.J., Schipper, A.M., 2018. Global patterns of current and future road infrastructure. *Environ. Res. Lett.* 13 (6), 064006. <https://doi.org/10.1088/1748-9326/aabd42>.
- Milcu, A.I., Hanspach, J., Abson, D., Fischer, J., 2013. Cultural ecosystem services: a literature review and prospects for future research. *Ecol. Soc.* 18 (3). <https://www.jstor.org/stable/26269377>.
- Millennium Ecosystem Assessment, 2005. *Ecosystems and Human Well-Being: Wetlands and Water*. World Resources Institute.
- Nabout, J.C., Tessarolo, G., Baptista Pinheiro, G.H., Matos Marquez, L.A., de Carvalho, R. A., 2022. Unraveling the paths of water as aquatic cultural services for the ecotourism in Brazilian protected areas. *GLOBAL ECOLOGY AND CONSERVATION* 33, e01958. <https://doi.org/10.1016/j.gecco.2021.e01958>.
- Nie, X., Wang, Q., Wu, J., Wang, H., Chen, Z., He, J., 2022. Effectively enhancing perceptions of cultural ecosystem services: a case study of a karst cultural ecosystem. *J. Environ. Manag.* 315, 115189. <https://doi.org/10.1016/j.jenvman.2022.115189>.
- Nyelele, C., Keske, C., Chung, M.G., Guo, H., Egoh, B.N., 2023. Using social media data and machine learning to map recreational ecosystem services. *Ecol. Indic.* 154, 110606. <https://doi.org/10.1016/j.ecolind.2023.110606>.
- Oteros-Rozas, E., Martín-López, B., Fagerholm, N., Bieling, C., Plieninger, T., 2018. Using social media photos to explore the relation between cultural ecosystem services and landscape features across five European sites. *Ecol. Indic.* 94, 74–86. <https://doi.org/10.1016/j.ecolind.2017.02.009>.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., et al., 2019. PyTorch: an imperative style, high-performance deep learning library (no. arXiv:1912.01703). arXiv. <https://doi.org/10.48550/arXiv.1912.01703>.

- Plieninger, T., Dijk, S., Oteros-Rozas, E., Bieling, C., 2013. Assessing, mapping, and quantifying cultural ecosystem services at community level. *Land Use Policy* 33, 118–129. <https://doi.org/10.1016/j.landusepol.2012.12.013>.
- Poe, M.R., Norman, K.C., Levin, P.S., 2014. Cultural dimensions of socioecological systems: key connections and guiding principles for conservation in coastal environments. *Conserv. Lett.* 7 (3), 166–175. <https://doi.org/10.1111/conl.12068>.
- Pouso, S., Uyarra, M.C., Borja, Á., 2018. The recovery of estuarine quality and the perceived increase of cultural ecosystem services by beach users: a case study from northern Spain. *J. Environ. Manag.* 212, 450–461. <https://doi.org/10.1016/j.jenvman.2018.02.033>.
- Prechelt, L., 1998. Early stopping—but when? In: Orr, G.B., Müller, K.-R. (Eds.), *Neural Networks: Tricks of the Trade*. Springer, pp. 55–69. https://doi.org/10.1007/3-540-49430-8_3.
- Richards, D.R., Tunçer, B., 2018. Using image recognition to automate assessment of cultural ecosystem services from social media photographs. *Ecosyst. Serv.* 31, 318–325. <https://doi.org/10.1016/j.ecoser.2017.09.004>.
- Ridding, L.E., Redhead, J.W., Oliver, T.H., Schmucki, R., McGinlay, J., Graves, A.R., Morris, J., Bradbury, R.B., King, H., Bullock, J.M., 2018. The importance of landscape characteristics for the delivery of cultural ecosystem services. *J. Environ. Manag.* 206, 1145–1154. <https://doi.org/10.1016/j.jenvman.2017.11.066>.
- Sayer, C.A., Fernando, E., Jimenez, R.R., Macfarlane, N.B.W., Rapacciuolo, G., Böhm, M., Brooks, T.M., Contreras-MacBeath, T., Cox, N.A., Harrison, I., Hoffmann, M., Jenkins, R., Smith, K.G., Vié, J.-C., Abbott, J.C., Allen, D.J., Allen, G.R., Barrios, V., Boudot, J.-P., et al., 2025. One-quarter of freshwater fauna threatened with extinction. *Nature* 638 (8049), 138–145. <https://doi.org/10.1038/s41586-024-08375-z>.
- Schirpke, U., Ghermandi, A., Sinclair, M., Van Berkel, D., Fox, N., Vargas, L., Willemen, L., 2023. Emerging technologies for assessing ecosystem services: a synthesis of opportunities and challenges. *Ecosyst. Serv.* 63, 101558. <https://doi.org/10.1016/j.ecoser.2023.101558>.
- Shorten, C., Khoshgoftaar, T.M., 2019. A survey on image data augmentation for deep learning. *J. Big Data* 6 (1), 60. <https://doi.org/10.1186/s40537-019-0197-0>.
- Sinclair, M., Mayer, M., Woltering, M., Ghermandi, A., 2020. Valuing nature-based recreation using a crowdsourced travel cost method: a comparison to onsite survey data and value transfer. *Ecosyst. Serv.* 45, 101165. <https://doi.org/10.1016/j.ecoser.2020.101165>.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15 (56), 1929–1958.
- Strahler, A.N., 1952. Hypsometric (area-altitude) analysis of erosional topography. *GSA Bulletin* 63 (11), 1117–1142. [https://doi.org/10.1130/0016-606\(1952\)63\[1117:HAAOETJ2.0.CO;2](https://doi.org/10.1130/0016-606(1952)63[1117:HAAOETJ2.0.CO;2).
- Strayer, D., 2010. Freshwater biodiversity conservation: recent progress and future challenges. *J. North Am. Benthol. Soc.* 29 (1), 189–197. <https://doi.org/10.1899/08-171>.
- Tatem, A.J., 2017. WorldPop, open data for spatial demography. *Sci. Data* 4 (1), 170004. <https://doi.org/10.1038/sdata.2017.4>.
- Tenkanen, H., Di Minin, E., Heikinheimo, V., Hausmann, A., Herbst, M., Kajala, L., Toivonen, T., 2017. Instagram, flickr, or Twitter: assessing the usability of social media data for visitor monitoring in protected areas. *Sci. Rep.* 7 (1), 17615. <https://doi.org/10.1038/s41598-017-18007-4>.
- Terraube, J., Fernández-Llamazares, Á., Cabeza, M., 2017. The role of protected areas in supporting human health: a call to broaden the assessment of conservation outcomes. *Curr. Opin. Environ. Sustain.* 25, 50–58. <https://doi.org/10.1016/j.cosust.2017.08.005>.
- Thiele, J., von Haaren, C., Albert, C., 2019. Are river landscapes outstanding in providing cultural ecosystem services? An indicator-based exploration in Germany. *Ecol. Indic.* 101, 31–40. <https://doi.org/10.1016/j.ecolind.2019.01.003>.
- Tickner, D., Opperman, J.J., Abell, R., Acreman, M., Arthington, A.H., Bunn, S.E., Cooke, S.J., Dalton, J., Darwall, W., Edwards, G., Harrison, I., Hughes, K., Jones, T., Leclère, D., Lynch, A.J., Leonard, P., McClain, M.E., Murruven, D., Olden, J.D., et al., 2020. Bending the curve of global freshwater biodiversity loss: an emergency recovery plan. *Bioscience* 70 (4), 330–342. <https://doi.org/10.1093/biosci/biaa002>.
- Tufekci, Z., 2014. Big questions for social media big data: representativeness, validity and other methodological pitfalls (no. arXiv:1403.7400). arXiv. <https://doi.org/10.48550/arXiv.1403.7400>.
- Völker, S., Kistemann, T., 2011. The impact of blue space on human health and well-being – salutogenetic health effects of inland surface waters: a review. *Int. J. Hyg. Environ. Health* 214 (6), 449–460. <https://doi.org/10.1016/j.ijheh.2011.05.001>.
- Walz, U., Stein, C., 2014. Indicators of hemeroby for the monitoring of landscapes in Germany. *J. Nat. Conserv.* 22 (3), 279–289. <https://doi.org/10.1016/j.jnc.2014.01.007>.
- Whitelaw, P.A., King, Brian E.M., Tolkach, D., 2014. Protected areas, conservation and tourism – financing the sustainable dream. *J. Sustain. Tourism* 22 (4), 584–603. <https://doi.org/10.1080/09669582.2013.873445>.
- Winder, S.G., Lee, H., Seo, B., Lia, E.H., Wood, S.A., 2022. An open-source image classifier for characterizing recreational activities across landscapes. *People Nat* 4, 1249–1262. <https://doi.org/10.1002/pan3.10382>.
- Wood, S.A., Guerry, A.D., Silver, J.M., Lacayo, M., 2013. Using social media to quantify nature-based tourism and recreation. *Sci. Rep.* 3 (1), 2976. <https://doi.org/10.1038/srep02976>.
- Xu, H., Zhao, G., Fagerholm, N., Primdahl, J., Plieninger, T., 2020. Participatory mapping of cultural ecosystem services for landscape corridor planning: a case study of the silk roads corridor in zhangye, China. *J. Environ. Manag.* 264, 110458. <https://doi.org/10.1016/j.jenvman.2020.110458>.
- Zhang, H., Cisse, M., Dauphin, Y.N., Lopez-Paz, D., 2018. Mixup: beyond empirical risk minimization (no. arXiv:1710.09412). arXiv. <https://doi.org/10.48550/arXiv.1710.09412>.
- Zhang, Z., Lu, M., Ji, S., Yu, H., Nie, C., 2021. Rich CNN features for water-body segmentation from very high resolution aerial and satellite imagery. *Remote Sens.* 13 (10), 10. <https://doi.org/10.3390/rs13101912>.
- Zhang, H., Huang, R., Zhang, Y., Buhalis, D., 2022. Cultural ecosystem services evaluation using geolocated social media data: a review. *Tour. Geogr.* 24 (4–5), 646–668. <https://doi.org/10.1080/14616688.2020.1801828>.
- Zhuang, J., Tang, T., Ding, Y., Tatikonda, S.C., Dvornek, N., Papademetris, X., Duncan, J., 2020. AdaBelief optimizer: adapting stepsizes by the belief in observed gradients. *Adv. Neural Inf. Process. Syst.* 33, 18795–18806.
- Zielstra, D., Hochmair, H.H., 2013. Positional accuracy analysis of flickr and panoramio images for selected world regions. *Journal of Spatial Science* 58 (2), 251–273. <https://doi.org/10.1080/14498596.2013.801331>.